

Beyond the Hype: Emerging Technologies and the Risks of Misperception

By Andrew Luke

Abstract: Emerging technologies do not have to work as advertised to create strategic instability. Decision makers only have to believe that they do. This reflection argues that alarm about quantum sensing and AI often targets the wrong problem: technological advancement itself, rather than the gap between claimed capability and verified performance. Drawing on scholarship on emerging technologies and military assessment, it examines exaggerated claims about Chinese quantum sensing alongside Israel's misplaced confidence in its "smart border." These cases reveal two sides of the same assessment problem: fear of an adversary's unproven capabilities and confidence in one's own inadequately tested defenses. Both can distort judgments about military effectiveness and security. The argument is not that technological competition is unimportant or that investment in research should stop. It is that investment alone cannot address a risk driven by perception. Independent technical red-teaming and adversary capability assessments are needed to separate capability from propaganda without turning skepticism into complacency. The unresolved challenge is how policymakers can question technological claims without dismissing the threats those claims may conceal.

Alarm about emerging technologies like quantum sensing and AI tends to rely on a flawed premise which equates hype to capability. A closer look shows this is not the case. Quantum sensing's most cited threats remain years from the battlefield and are often propagated by the very states with an interest in appearing more advanced than they actually are. This does not suggest that the alarm is misplaced, rather it's aimed at the wrong target. Technology isn't the destabilizing force here; it's that deterrence and military assessment operate on perceived capability, and perception is often manipulated, exaggerated, or fails. That problem is more dangerous than the maximalist narrative, 'this changes everything' because it means that instability can occur even in the absence of a working weapon. It also means that the fix is not an investment in more fear, but better tools for separating capability from propaganda.

Alarmism creates a bipolar debate. The policy side contends that the speed at which AI innovation, quantum computing, and networked systems move represents an extraordinary discontinuity in military capabilities, one serious enough to erode long-held U.S. advantages. Academics, however, advocate for caution. Sechser, Narang, and Talmadge argue that "very few technologies fundamentally reshape the dynamics of international conflict," noting that even chemical weapons expected to transform warfare after their first use in the first World War, "quickly turned out to be less practical, easier to counter, and less effective than conventional high-explosives" (Sechser, Narang, & Talmadge, pp.728-729). Gamberini and Rubin extend this skepticism specifically to quantum sensing, arguing that public policy discourse should be evaluated "with a healthy skepticism" given the field's low technology readiness level (TRL) and the tendency of both media and interested states to overstate progress (Gamberini & Rubin, p.358).

Gamberini and Rubin provide a concrete example of hype not matching reality citing a case where the Chinese over promised and underdelivered the "next generation of their quantum radar system" designed to "detect stealth bombers" and "effectively monitor high-speed flying objects in the upper atmosphere and above" (Gamberini & Rubin, p. 360). They point to the third leg of the United States' Nuclear Triad, the U.S. Navy's Ohio Class ballistic missile submarine (SSBN). The SSBN's stealth and nuclear warhead delivery system would be at risk if the quantum radar system functioned now and not at some indeterminate time in the future. Gamberini and Rubin reinforce their point that "deterrence is based on perception" (Gamberini & Rubin, p. 362) and that even exaggerated or unrealized adversary sensing can erode confidence in a state's second-strike capability.

Another example is Israel's "smart border" and "smart fence" which used sophisticated technology, such as, "sensors, drones, automated weapons, robots, and cameras" and "a system of integrated sensors and concrete barriers" (Murphy & Rubin, p. 165) to monitor their border. Hamas' "nearly 4,000 fighters" defeated these "smart" systems using basic technology. The invasion ultimately cost "more than 40,000 Palestinians, most of them civilians" (Murphy & Rubin, p. 165) their lives. Murphy and Rubin illustrate Elsa Kania's observation

that AI's "disparate and qualitative" effects on military capability create uncertainty such that "the military balance may be merely perceived rather than adequately evaluated" (Murphy & Rubin, p. 162).

These examples demonstrate that the danger does not require the underlying technology to function as claimed. The only requirement is that decision makers act as though they do. Reframed this way, the speed of technological advancement does not fully discredit the case for concern. Despite Gamberini and Rubin's near-term capability skepticism, they still recommend that quantum sensing "remain a priority focus area for the Defense Department's R&D investment" (Gamberini & Rubin, p.367). They further emphasize the critical nature of maintaining the "competitive edge" in quantum sciences and not "cede technological advantage[s] to [our] competitors" (Gamberini & Rubin, p.367). The argument is not that the stakes are low, but that the timeline and the mechanism of risk are misunderstood.

The practical implication is that strategic stability cannot be managed through capability competition alone. If perception, not verified capability drives deterrence confidence, procurement decisions, and crisis behavior, then investment in R&D will not be enough. Policy makers will need mechanisms for assessing and countering propaganda and misperception, such as independent technical red-teaming and adversary capability assessments in a deterrence framework. Israel's confidence in the assessment of their own "smart fence" and "smart border" created a blind spot. This raises questions worth pursuing: does treating an adversary's claim with skepticism risk the same complacency that undermined Israel's border defense? And is it realistic to believe that there are any mechanisms policy makers can implement to accurately assess whether claims about the state of their technological innovations are true?

References:

Gamberini, S. J., & Rubin, L. (2021). Quantum sensing's potential impacts on strategic deterrence and modern warfare. *Orbis*, 65(2), 354–368.

Murphy, D., & Rubin, L. (2025). The fourth industrial revolution and international security. *Survival*, 67(1), 159–172. <https://doi.org/10.1080/00396338.2025.2459023>

Sechser, T. S., Narang, N., & Talmadge, C. (2019). Emerging technologies and strategic stability in peacetime, crisis, and war. *Journal of Strategic Studies*, 42(6), 727–735.